

Melde's apparatus

1 Aim

To determine the frequency of an electrically maintained tuning fork by Melde's experiment.

2 Introduction

This experiment uses an electrically maintained tuning fork to set in standing waves in a string. One end of this string is tied to one of the prongs of the tuning fork and the second end of this string passes over a pulley in such a way that a small pan can be hang on it. Small weights are put in the pan to create tension in the string.

The transverse vibrations are produced in the string and by adjusting the length and/or weights in the pan, these vibrations are reflected from the point of contact between the string and the pulley and a number of loops appear on the string, as shown in the picture. When the prongs of the tuning fork vibrate in a plane, parallel to the direction of propagation of the waves, it is called the longitudinal position. If the tuning fork, along with the wooden block is rotated through 90° , so that the plane of vibration of the tuning fork is normal (i.e. at right angles) to the direction of propagation, it is called the transverse position. For the same length and same weight in the pan (i.e. for the same tension) the number of loops in the transverse position is double than in the longitudinal position.

- Transverse Mode:

In this mode, the time of completion of one vibration of the tuning fork and the time for the string to complete one vibration is the same. In this case, the frequency is given by

$$\nu = \frac{n}{2l} \sqrt{\frac{T}{\mu}},$$

where n is the number of loops, T is the tension and μ is mass per unit length of the string.

- Longitudinal mode:

In this arrangement, the vibrations of the prongs of the tuning fork are parallel to the length of the string. In this Case, the frequency of the thread is half of that of the tuning fork and therefore the relation between the number of loops and the frequency is given by

$$\nu = \frac{n}{l} \sqrt{\frac{T}{\mu}},$$

Electrically maintained tuning fork:

The stem of a (large) tuning fork is clamped rigidly. A short metal strip is attached to one of the prongs and a screw maintains contact with the metal strip. An electromagnet is placed between the prongs and one end of the coil is connected to the screw and the other is connected to a battery/electrical supply and then connects to the stem of the tuning fork (see the apparatus carefully). When the screw touches the metal strip, the circuit is closed and the tuning fork begins to vibrate.

3 Apparatus in the lab and procedure

1. Place a few weights in the pan and switch on the tuning fork mechanism.
2. Adjust the position of the pulley such that the string has resonant vibrations and well defined loops.
3. Compute the length of each node (divide the length by number of nodes) and compute tension.
4. Repeat the experiment with different weights.
5. Measure a metre length of the thread and weight it to find mass per unit length.

4 Precautions

1. The thread should be as uniform as possible. You can ask the lab staff to help replace the string if it is frayed.
2. Make sure that the pan with weights is not oscillating.



Figure 1: Apparatus in the lab and a view of the stationary waves.